Unit outline_

DATA5207: Data Analysis in the Social Sciences

Semester 1, 2026 [Normal day] - Camperdown/Darlington, Sydney

On this page
- Overview
- Assessment
- Learning support
- Weekly schedule
- Learning outcomes
- Responding to student feedback
- Additional information

Useful links
- Student Portal →
- Timetables →
- Key dates →
- Canvas →
- Student Admin →
- Health management →

Overview

Data science is a new, rapidly expanding field. There is an unprecedented demand from technology companies, financial services, government and not-for-profits for graduates who can effectively analyse data. This subject will help students gain a critical understanding of the strengths and weaknesses of quantitative research, and acquire practical skills using different methods and tools to answer relevant social science questions. This subject will offer a nuanced combination of real-world applications to data science methodology, bringing an awareness of how to solve actual social problems to the Master of Data Science. We cover topics including elections, criminology, economics and the media. You will clean, process, model and make meaningful visualisations using data from these fields, and test hypotheses to draw inferences about the social world. Techniques covered range from descriptive statistics and linear and logistic regression, the analysis of data from randomised experiments, model selection for prediction and classification tasks, to the analysis of unstructured text as data, multilevel and geospatial modelling, all using the open source program R. In doing this, not only will we build on the skills you have already mastered through this degree, but explore different ways to use them once you graduate.

Unit details and rules

Academic unit	Computer Science
Credit points	6
Prerequisites ?	None
Corequisites ?	None
Prohibitions ?	DATA4207
Assumed knowledge ?	None
Available to study abroad and exchange students	Yes

Teaching staff

Coordinator	Kathryn Robison, kathryn.robison@sydney.edu.au

Assessment

ⓘ The census date for this unit availability is **31 March 2026**

Details | Criteria

Type	Description	Weight	Due	Length	Use of AI
Written work 🚧	**Research project** Complete major research project.	50%	Formal exam period **Due date**: 08 Mar 2026 at 23:59 **Closing date**: 22 Jun 2026	3500 words	AI allowed
Outcomes assessed: LO2 LO3 LO4 LO5 LO7					
In-person practical, skills, or performance task or test	**Lecture Quiz - in class** in-class secure quizzes on lecture material (10 total)	10%	Multiple weeks	10 x 3 minute quiz (30 mins total)	AI prohibited
Outcomes assessed: LO1 LO3 LO4 LO5					
Contribution	**Muddy Cards** Questions to promote synthesis of concepts	4%	Multiple weeks	N/A	AI allowed
Outcomes assessed: LO1 LO3 LO4 LO5					
Research analysis 🚧	**Research plan** Develop plan for major research project.	20%	Week 05 **Due date**: 27 Mar 2026 at 23:59 **Closing date**: 10 Apr 2026	-	AI allowed
Outcomes assessed: LO2 LO3 LO4 LO5 LO7					
Practical skill 👥	**Group work 1** Group work.	8%	Week 08 **Due date**: 20 Apr 2026 at 23:59 **Closing date**: 04 May 2026	variable	AI allowed
Outcomes assessed: LO1 LO2 LO3 LO4 LO5 LO6 LO7					
Practical skill 👥	**Group work 2** group work	8%	Week 10 **Due date**: 04 May 2026 at 23:59 **Closing date**: 18 May 2026	variable	AI allowed
Outcomes assessed: LO5 LO6 LO7					

🚧 = hurdle task ? 👥 = group assignment ?

Assessment summary

Use of generative artificial intelligence (AI)

You can use generative AI tools for open assessments. Restrictions on AI use apply to secure, supervised assessments used to confirm if students have met specific learning outcomes.

Refer to the assessment table above to see if AI is allowed, for assessments in this unit and check Canvas for full instructions on assessment tasks and AI use.

If you use AI, you must always acknowledge it. Misusing AI may lead to a breach of the Academic Integrity Policy.

Visit the Current Students website for more information on AI in assessments, including details on how to acknowledge its use.

Late submission

In accordance with University policy, these penalties apply when written work is submitted after 11:59pm on the due date:

- Deduction of **5%** of the maximum mark for each calendar day after the due date.
- After ten calendar days late, a mark of zero will be awarded.

Academic integrity

The University expects students to act ethically and honestly and will treat all allegations of academic integrity breaches seriously.

Our website provides information on academic integrity and the resources available to all students. This includes advice on how to avoid common breaches of academic integrity. Ensure that you have completed the Academic Honesty Education Module (AHEM) which is mandatory for all commencing coursework students

Penalties for serious breaches can significantly impact your studies and your career after graduation. It is important that you speak with your unit coordinator if you need help with completing assessments.

Visit the Current Students website for more information on AI in assessments, including details on how to acknowledge its use.

Learning support

Simple extensions

If you encounter a problem submitting your work on time, you may be able to apply for an extension of five calendar days through a simple extension. The application process will be different depending on the type of assessment and extensions cannot be granted for some assessment types like exams.

Special consideration

If exceptional circumstances mean you can't complete an assessment, you need consideration for a longer period of time, or if you have essential commitments which impact your performance in an assessment, you may be eligible for special consideration or special arrangements.

Special consideration applications will not be affected by a simple extension application.

Using AI responsibly

Co-created with students, AI in Education includes lots of helpful examples of how students use generative AI tools to support their learning. It explains how generative AI works, the different tools available and how to use them responsibly and productively.

Support for students

The Support for Students Policy reflects the University's commitment to supporting students in their academic journey and making the University safe for students. It is important that you read and understand this policy so that you are familiar with the range of support services available to you and understand how to engage with them.

The University uses email as its primary source of communication with students who need support under the Support for Students Policy. Make sure you check your University email regularly and respond to any communications received from the University.

Learning resources and detailed information about weekly assessment and learning activities can be accessed via Canvas. It is essential that you visit your unit of study Canvas site to ensure you are up to date with all of your tasks.

If you are having difficulties completing your studies, or are feeling unsure about your progress, we are here to help. You can access the support services offered by the University at any time:

Support and Services (including health and wellbeing services, financial support and learning support)
Course planning and administration
Meet with an Academic Adviser

Weekly schedule

WK	Topic	Learning activity	Learning outcomes
Week 01	Understanding the social world using data science	Lecture (1 hr)	LO1
Week 02	Visualising social science data	Lecture (1 hr)	LO1 LO2
Week 03	Human behaviour studies and their errors	Lecture (1 hr)	LO1 LO2
Week 04	Surveys and probabilities	Lecture (1 hr)	LO1 LO2 LO3
Week 05	Predicting outcomes in the social world	Lecture (1 hr)	LO1 LO2 LO3 LO4
Week 06	Writing about social science data (I)	Lecture (1 hr)	LO1 LO2 LO3 LO5 LO4
Week 07	Measuring latent variables in the social world	Lecture (1 hr)	LO1 LO2 LO3 LO5 LO4
Week 08	Validating social science data	Lecture (1 hr)	LO1 LO2 LO3 LO5 LO4
Week 09	Using spatial data to understand the world	Lecture (1 hr)	LO1 LO2 LO3 LO5 LO4
Week 10	Writing about social science data (II)	Lecture (1 hr)	LO1 LO2 LO3 LO5 LO4
Week 11	Data journalism	Lecture (1 hr)	LO1 LO2 LO3 LO5 LO4
Week 12	Communication analyses	Lecture (1 hr)	LO1 LO2 LO3 LO5 LO4
Week 13	Conclusion + developing a research project	Lecture (1 hr)	LO1 LO6 LO7 LO2 LO3 LO5 LO4
Multiple weeks	practical application of lecture concepts	Tutorial (13 hr)	LO1 LO6 LO7 LO2 LO3 LO5 LO4

Attendance and class requirements

COVID-19 Announcement:

This unit will be taught as a combined face-to-face and online class.

You will have the option to enrol in either online or in-person lectures and labs. Online lectures and labs will all be run live over Zoom.

Recordings will be made available to students and accessibility needs will be considered.

Study commitment

Typically, there is a minimum expectation of 1.5-2 hours of student effort per week per credit point for units of study offered over a full semester. For a 6 credit point unit, this equates to roughly 120-150 hours of student effort in total.

Required readings

These will be available through the unit Canvas page.

Learning outcomes

Learning outcomes are what students know, understand and are able to do on completion of a unit of study. They are aligned with the University's graduate qualities and are assessed as part of the curriculum.

Outcomes | Graduate qualities

At the completion of this unit, you should be able to:

LO1. demonstrate familiarity with the various ethical issues and professional standards around the gathering of data
LO2. autonomously collect, collate, assess and compare data from multiple sources, such as the Australian Bureau of Statistics and the Australian Data Archive. You will be able to discern the quality of data to a minute level, and be able to draw a broad range of insights from data of various degrees of statistical significance
LO3. apply established data analytical methodology in a sophisticated manner using R and have a medium degree of proficiency in methodological procedures to approach complex problems specifically related to the social sciences
LO4. apply a theoretical understanding of statistical methods to practical problems around data gathering methodology, statistical significance and sample sizing, and autonomously create basic design frameworks for statistical modelling problems.
LO5. utilise industry-leading concepts and frameworks in your pedagogy and direct formidable amounts of data for protracted, complex insights into areas such as polling data and demography
LO6. demonstrate proficiency in the delivery of a small-scale project, and the management of the project from initial conception to delivery to evaluation
LO7. present data and reports of a high standard using RStudio

Responding to student feedback

This section outlines changes made to this unit following staff and student reviews.

This unit has been offered for a number of years and the UoS coordinators have continually worked to improve the quality of teaching materials, learning activities and forms of assessment based on student feedback.

Additional information

Canvas

Details about weekly topics and assessments will be available on the unit Canvas page.

Late submissions

University policies on late submissions apply.

Academic Dishonesty and Plagiarism

All written assignments submitted in this unit of study will be submitted to the similarity detecting software program known as Turnitin. Turnitin searches for matches between text in your written assessment task and text sourced from the Internet, published works and assignments that have previously been submitted to Turnitin for analysis.

There will always be some degree of text-matching when using Turnitin. Text-matching may occur in use of direct quotations, technical terms and phrases, or the listing of bibliographic material. This does not mean you will automatically be accused of academic dishonesty or plagiarism, although Turnitin reports may be used as evidence in academic dishonesty and plagiarism decision-making processes.

Computer programming assignments may also be checked by specialist code similarity detection software. The Faculty of Engineering currently uses the MOSS similarity detection engine (see http://theory.stanford.edu/~aiken/moss/) . These programs work in a similar way to TII in that they check for similarity against a database of previously submitted assignments and code available on the internet, but they have added functionality to detect cases of similarity of holistic code structure in cases such as global search and replace of variable names, reordering of lines, changing of comment lines, and the use of white space.

IMPORTANT: School policy relating to Academic Dishonesty and Plagiarism.

In assessing a piece of submitted work, the School of Computer Science may reproduce it entirely, may provide a copy to another member of faculty, and/or to an external plagiarism checking service or in-house computer program and may also maintain a copy of the assignment for future checking purposes and/or allow an external service to do so.

Disclaimer

Important: the University of Sydney regularly reviews units of study and reserves the right to change the units of study available annually. To stay up to date on available study options, including unit of study details and availability, refer to the relevant handbook.

The information included on this page is correct at 05 Feb 2026.

To help you understand common terms that we use at the University, we offer an online glossary.

We recognise and pay respect to the Elders and communities - past, present, and emerging - of the lands that the University of Sydney's campuses stand on. For thousands of years they have shared and exchanged knowledges across innumerable generations for the benefit of all.

Learn more

THE UNIVERSITY OF SYDNEY

Media
- News
- Find an expert
- Media contacts

Student links
- How to log in to University systems
- Key dates
- Class timetables
- Policy

About us
- Our world university rankings
- Faculties and schools
- Research centres
- Campus locations

Connect
- Contact us
- Find a staff member
- Careers at Sydney
- Emergencies and personal safety
- Feedback, concerns and complaints

GROUP OF EIGHT AUSTRALIA APRU Athena SWAN

Disclaimer Privacy statement Accessibility Website feedback Your privacy settings

ABN: 15 211 513 464
CRICOS Number: 00026A
TEQSA: PRV12057